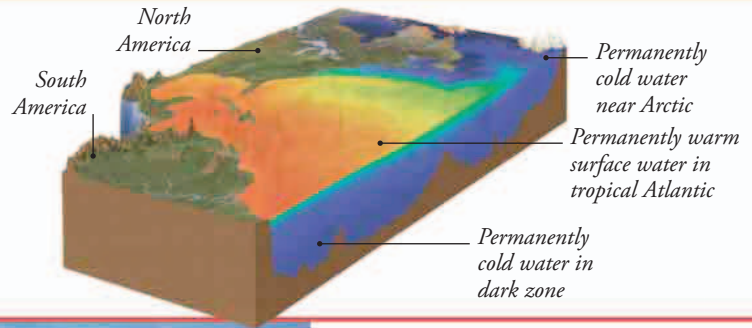
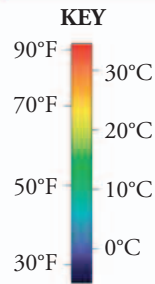


## THERMOCLINE

Tropical oceans are warm at the surface, reaching 86°F (30°C). But the temperature falls rapidly with depth to 39°F (4°C) in the twilight zone, and almost freezing in the dark zone. In the tropics the warm surface water rarely mixes with the colder water below it, and the boundary between the two is called the thermocline.



Sooty shearwater

Anchovies

Common dolphins

Blue shark

Comb jellyfish

Hatchetfish

Anglerfish

Vampire squid

Deep-sea squid

Red shrimp



### ▲ CLEAR BLUE WATER

In open tropical oceans the thermocline usually stops dissolved minerals reaching the sunlit surface water, where they would encourage the growth of microscopic plankton. As a result, there is very little plankton in most tropical seas. This is why the water is crystal clear. In cooler seas, the thermocline breaks down in winter, allowing nutrients to fuel plankton growth.



### ▲ SPECIALLY ADAPTED

Ocean water contains a lot of oxygen, which is vital to animal life. The colder the water, the more oxygen it has. But deep in the twilight zone, there is a region where most of the oxygen has been used up by bacteria feeding on dead plankton sinking from above. The only animals that can survive in this region are specially adapted creatures like this vampire squid.

